



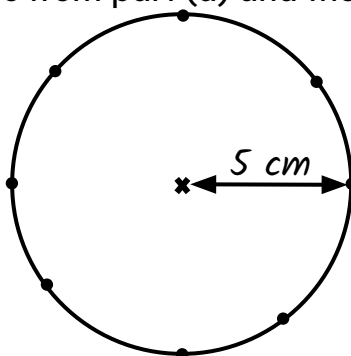
Understanding constructing a circle

Task A: Properties of a circle

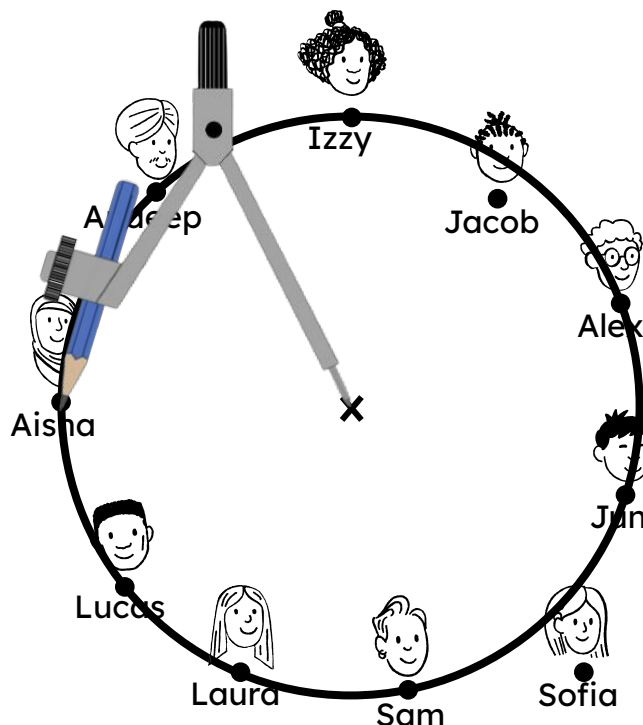
- 1) You will need a blank piece of A4 paper for question 1. It will not fit on this sheet. Mark a cross in the middle of your plain piece of paper.
- a) Mark 8 points that are exactly 7 cm away from the cross.
- b) Draw a circle with radius 7 cm and with the centre at the cross.
- c) What do you notice about the points from part (a) and the circle from part (b)?

E.g. Not drawn to scale.

All of the points lie on the circle.



- 2) Aisha marks a cross on a piece of paper and marks a point exactly 5 units away from it. She asks everyone else to mark a point 5 units away from the cross.



- a) How could you use a pair of compasses to check whether everyone's point is the correct distance away from the cross?

Draw a circle with the cross at the centre and Aisha's point on the circle.

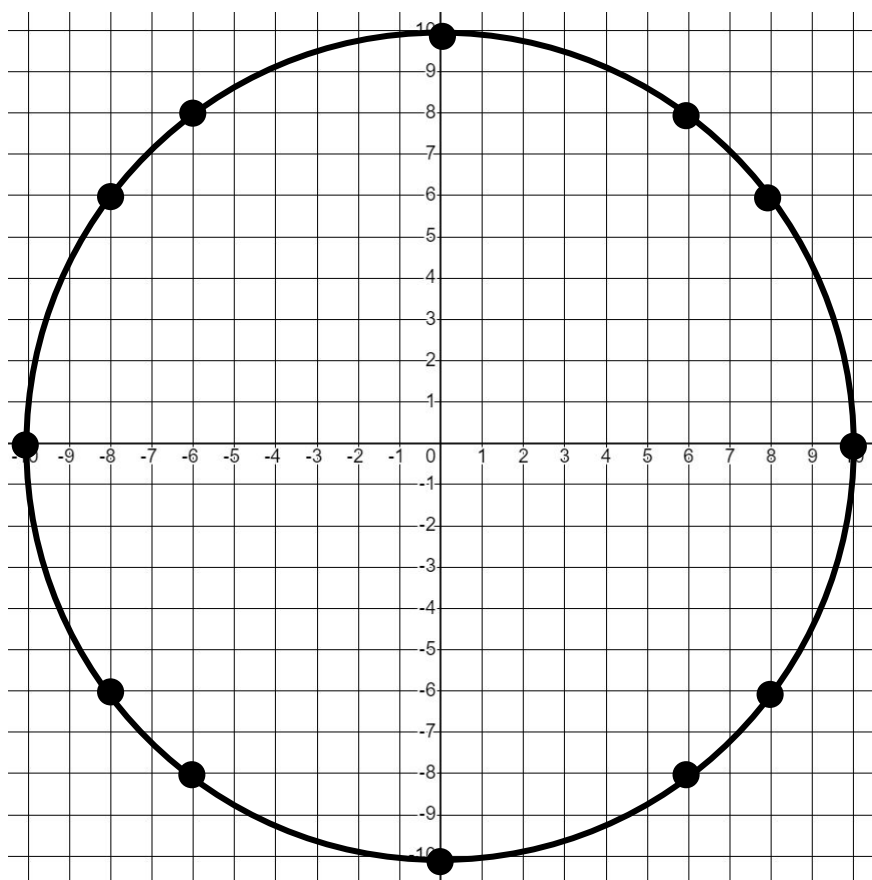
- b) Whose point is too close to the cross?
- c) Whose point is too far from the cross?

Jacob

Sofia



3) Find the coordinates of 12 points which are exactly 10 units from the origin.



$(10, 0)$

$(0, 10)$

$(-10, 0)$

$(0, -10)$

$(8, 6)$ $(-8, -6)$

$(6, 8)$ $(-6, -8)$

$(-6, 8)$ $(6, -8)$

$(-8, 6)$ $(8, -6)$

These are the only coordinates with integer values. There are more coordinates with decimal values.

Task B: Properties of an arc of a circle

1) W, X, Y and Z are arcs from four circles.

A, B, C and D are four possible centres for these circles.

Match the arcs to their centres.

A and Y

B and X

C and W

D and Z

